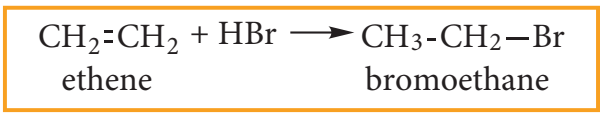
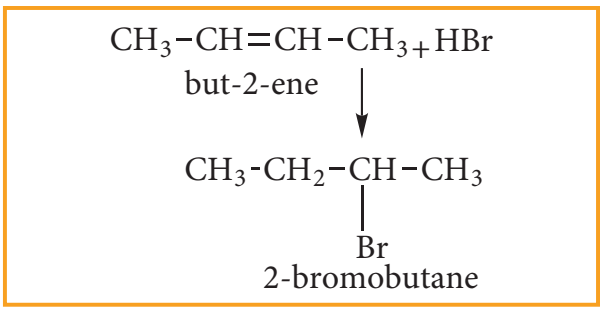
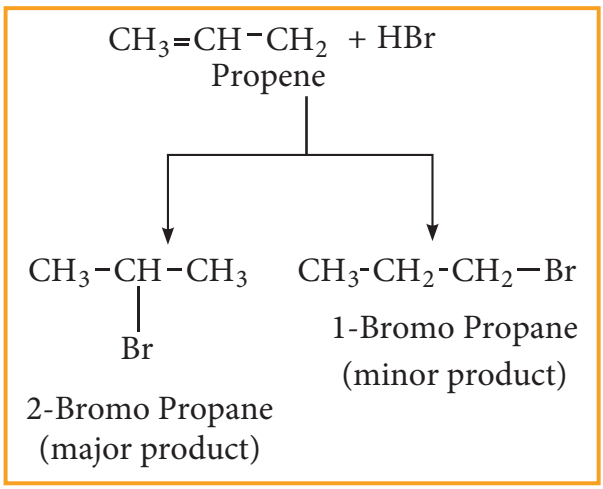
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(b) Addition HBr to unsymmetrical alkene:

In the addition of hydrogen halide to an unsymmetrical alkene, two products are obtained.

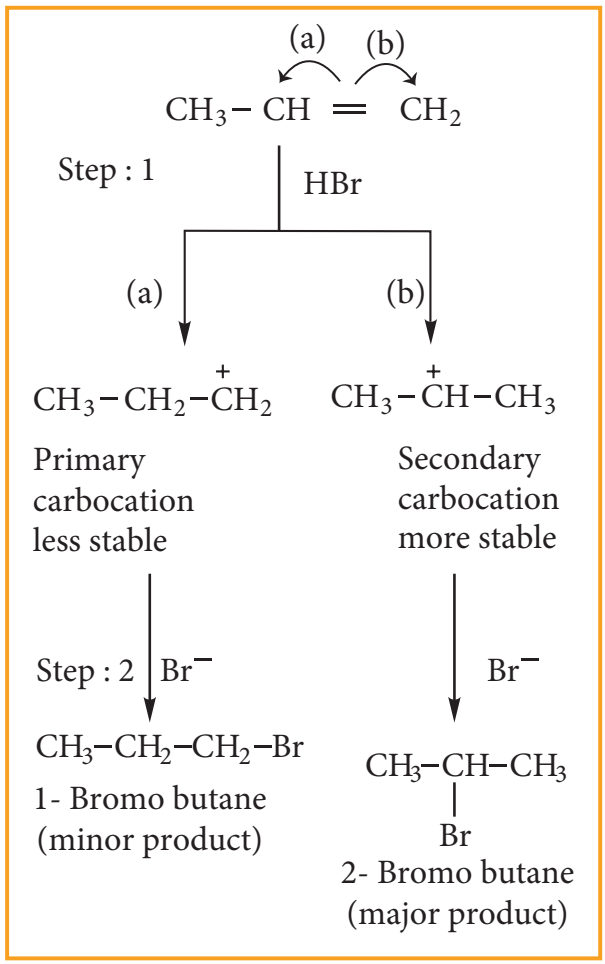


Mechanism:

Consider addition of HBr to propene

Step: 1 Formation of electrophile:

In H-Br, Br is more electronegative than H. When bonded electron moves toward Br, polarity is developed and creates an electrophile H^+ which attaches the double bond to form carbocation, as shown below.



Step:2 Secondary carbocation is more stable than primary carbocation and it predominates over a the primary carbocation.

Step:3 The Br^- ion attack the 2° carbocation to from 2-Bromobutane, the major product.

Consider addition of HBr to 3-methyl-1-butene. Here the expected product according to markovnikoff's rule is 2-bromo-3-methyl butane but the actual major product is 2-Bromo-2-methyl butane. This is because, the secondary carbocation formed during the reaction rearranged to more stable tertiary carbocation. Attack of Br^- on this tertiary carbocation gives the major product 2-bromo-2-methyl butane.